

HAFIZ MUHAMMAD

AI & Backend Engineer | Python, FastAPI, Django, LangChain & LangGraph, React

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PROFESSIONAL SUMMARY

AI & Backend Engineer with 1+ years shipping production Python web apps, RESTful APIs, and agentic AI systems using LangChain and LangGraph. Skilled in RAG pipelines, vector databases (ChromaDB), and real-time AI voice/chat systems, with full-stack delivery experience across Django, FastAPI, Flask, and React - from schema design to Linux deployment.

TECHNICAL SKILLS

AI & Agentic Systems: LangChain, LangGraph, RAG Pipelines, Prompt Engineering, ChromaDB, Groq API, Ollama, Whisper AI, LLM Integration

Backend & APIs: Python, FastAPI, Django, Flask, RESTful API Design, MVC Architecture, OOP, JWT Auth, WebSockets

Frontend & DB: React, JavaScript, HTML5/CSS3, MySQL, SQLAlchemy, Django ORM, ChromaDB (Vector DB)

Tools & Deployment: Git, GitHub, Linux Server Deployment, Stripe API, Telnix API

Certifications: Databases & SQL for Data Science (IBM) · OOP in Java (Oracle) · Design Patterns (FreeCodeCamp) · Linux Management (Linux Foundation)

PROFESSIONAL EXPERIENCE

AI & Backend Engineer - PureLogics, Lahore

July 2025 – Present

- Build and ship production Python web applications using Django and FastAPI with MVC architecture, documenting RESTful API endpoints that frontend teams integrate directly against.
- Independently architect and deliver AI service modules and LangChain/LangGraph-powered agentic pipelines used in production features.
- Administer MySQL databases (schema design, query optimization) and deploy applications to Linux servers, ensuring reliable uptime.
- Participate in structured code reviews, improving code quality across the team's Python codebase.

Lab Instructor - TopSkills, Lahore

2026

- Deliver hands-on lab sessions on Python and applied AI/agentic systems, translating real-world production experience into practical, project-based lessons for students.
- Mentor students through hands-on coding exercises and project work, building their debugging skills and confidence with core programming and AI concepts.

KEY PROJECTS

EnrollAI - AI-Powered Student Enrollment SaaS - AI & Backend Developer

2026

- Architected a self-hosted, multi-tenant SaaS platform with AI agents (LangChain + LangGraph) that hold natural conversations, answer course-specific questions via RAG, and grade uploaded CVs for course eligibility.
- Built a RAG pipeline (PyPDF + HuggingFace Sentence Transformers + ChromaDB) so the agent retrieves exact course details and institute rules from uploaded prospectuses, and integrated Groq for low-latency LLM inference.
- Delivered an omnichannel intake experience combining a white-labeled WebSocket chat widget with Meta WhatsApp Business API automation, plus a Flask/MySQL/Redis backend with per-feature AI usage and cost telemetry.

Stack: Python, Flask, LangChain, LangGraph, ChromaDB, Groq, PyPDF, HuggingFace, WhatsApp API, WebSockets, MySQL, Redis, Git

CallerAI - AI-Powered Cold Calling Platform - Full Stack Developer

2025

- Engineered a real-time AI voice agent that autonomously handles outbound sales calls, orchestrating Whisper AI (speech-to-text) to LLaMa 3.1 via Groq (reasoning) to Cartesia Sonic-2 (voice synthesis) through LangGraph, streamed live over WebSockets with sub-second latency.
- Built a private RAG knowledge base with LangChain + ChromaDB so the agent answers only from approved business/product data and deflects off-topic queries, reducing hallucination risk.

- Integrated Telnyx for live PSTN calls plus a browser-based web-call mode, with automated post-call analytics (summary, sentiment score, outcome label) on every call.

Stack: Python, FastAPI, LangChain, LangGraph, ChromaDB, Whisper AI, Groq, Telnyx, WebSockets, MySQL, React, Git

VoiceFlow - AI-Powered Voice Dictation System - Solo Developer

2026

- Engineered a global voice-to-text dictation agent in Python that intercepts system-wide hotkeys to capture audio and inject transcribed text into any active application with sub-second latency.
- Integrated offline speech-to-text using Faster-Whisper for high-accuracy, privacy-focused local transcription, plus an on-the-fly AI grammar-polish pipeline via Groq (LLaMA 3) or local Ollama.
- Built a multi-process desktop dashboard (Flask + Pywebview) for hotkey and API configuration, and packaged the app as a standalone Windows executable using PyInstaller and Inno Setup.

Stack: Python, Flask, Faster-Whisper, Groq, Ollama, PyAudio, Pywebview, PyInstaller, Git

Live: <https://hafizmuhammadshah42895.github.io/VoiceFlow/>

Premium Matrimony - AI Matchmaking Platform - Full Stack Developer

2025

- Built an AI-powered match-recommendation engine using LangChain + ChromaDB vector embeddings that surfaces the top 5 most compatible profiles per user with plain-English compatibility explanations.
- Designed a feedback learning loop where user likes/skips update individual preference profiles, improving the relevance of future match suggestions over time.
- Implemented real-time chat (WebSockets), OTP email verification, JWT-based authentication, and a full admin panel for user and CNIC verification management.

Stack: Python, Django, FastAPI, LangChain, ChromaDB, WebSockets, MySQL, React, JWT, REST APIs, Git

EDUCATION

B.Sc. Computer Science - Riphah International University